Visualize the data using Python libraries matplotlib, seaborn by plotting the graphs on Air Quality Dataset

```
# import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Reading csv file
df = pd.read_csv('Air Quality.csv')

df.head()
```

```
         Date      Time  CO(GT)  PT08.S1(CO)  NMHC(GT)  C6H6(GT)  \
0  10/03/2004  18:00:00     2.6       1360.0     150.0      11.9
1  10/03/2004  19:00:00     2.0       1292.0     112.0       9.4
2  10/03/2004  20:00:00     2.2       1402.0      88.0       9.0
3  10/03/2004  21:00:00     2.2       1376.0      80.0       9.2
4  10/03/2004  22:00:00     1.6       1272.0      51.0       6.5

   PT08.S2(NMHC)  NOx(GT)  PT08.S3(NOx)  NO2(GT)  PT08.S4(NO2)
PT08.S5(O3)  \
0         1046.0    166.0        1056.0    113.0        1692.0
1268.0
1          955.0    103.0        1174.0     92.0        1559.0
972.0
2          939.0    131.0        1140.0    114.0        1555.0
1074.0
3          948.0    172.0        1092.0    122.0        1584.0
1203.0
4          836.0    131.0        1205.0    116.0        1490.0
1110.0

      T    RH      AH  Unnamed: 15  Unnamed: 16
0  13.6  48.9  0.7578          NaN          NaN
1  13.3  47.7  0.7255          NaN          NaN
2  11.9  54.0  0.7502          NaN          NaN
3  11.0  60.0  0.7867          NaN          NaN
4  11.2  59.6  0.7888          NaN          NaN
```

```
# Check info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9471 entries, 0 to 9470
Data columns (total 17 columns):
 #   Column         Non-Null Count  Dtype
```

```
---   ------          --------------  -----
 0   Date            9357 non-null   object
 1   Time            9357 non-null   object
 2   CO(GT)          9357 non-null   float64
 3   PT08.S1(CO)     9357 non-null   float64
 4   NMHC(GT)        9357 non-null   float64
 5   C6H6(GT)        9357 non-null   float64
 6   PT08.S2(NMHC)   9357 non-null   float64
 7   NOx(GT)         9357 non-null   float64
 8   PT08.S3(NOx)    9357 non-null   float64
 9   NO2(GT)         9357 non-null   float64
 10  PT08.S4(NO2)    9357 non-null   float64
 11  PT08.S5(O3)     9357 non-null   float64
 12  T               9357 non-null   float64
 13  RH              9357 non-null   float64
 14  AH              9357 non-null   float64
 15  Unnamed: 15     0 non-null      float64
 16  Unnamed: 16     0 non-null      float64
dtypes: float64(15), object(2)
memory usage: 1.2+ MB
```

```
# Check missing values
df.isnull().sum()
```

```
Date               114
Time               114
CO(GT)             114
PT08.S1(CO)        114
NMHC(GT)           114
C6H6(GT)           114
PT08.S2(NMHC)      114
NOx(GT)            114
PT08.S3(NOx)       114
NO2(GT)            114
PT08.S4(NO2)       114
PT08.S5(O3)        114
T                  114
RH                 114
AH                 114
Unnamed: 15       9471
Unnamed: 16       9471
dtype: int64
```

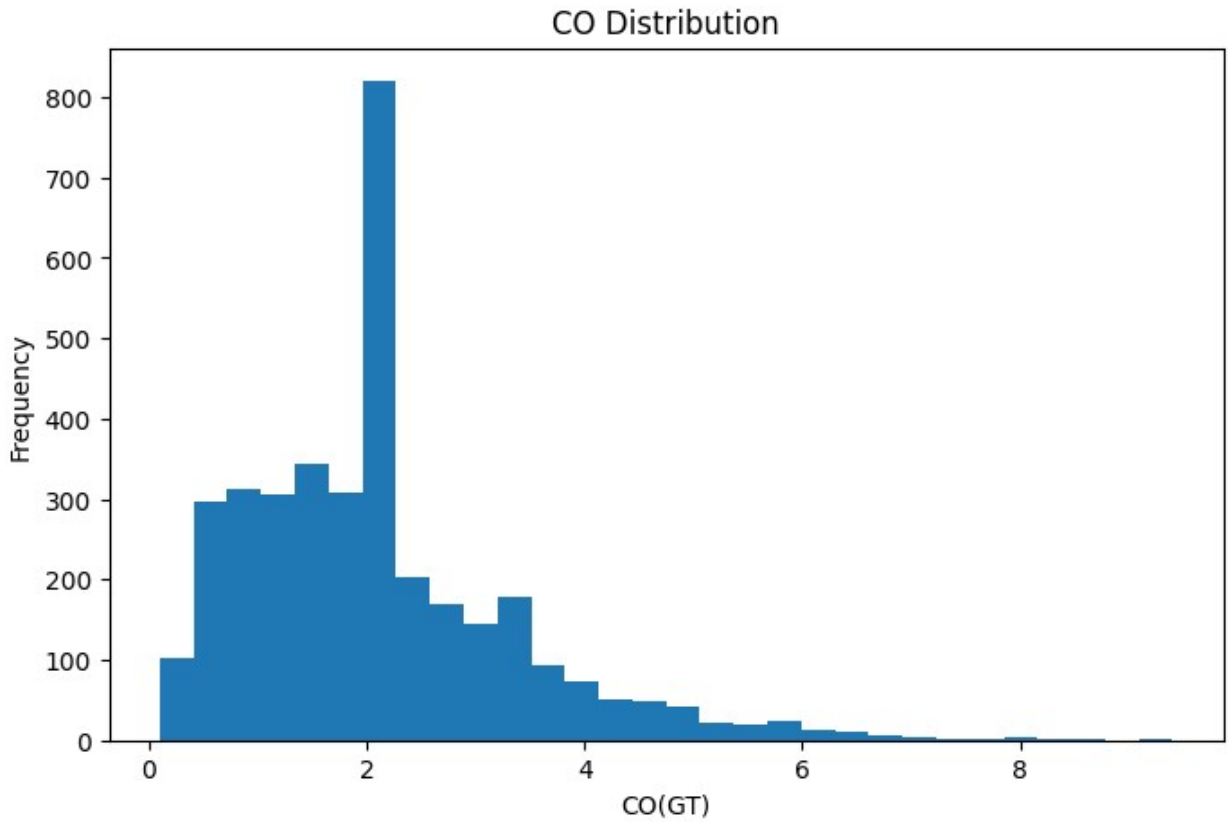
```
# Drop unwanted columns
df.drop(columns=['Unnamed: 15', 'Unnamed: 16'], inplace=True)

# Replace invalid values (-200 means missing sensor data)
df.replace(-200, np.nan, inplace=True)

# Convert Date + Time into Datetime
```

```
df['Datetime'] = pd.to_datetime(df['Date'] + ' ' + df['Time'],
errors='coerce')

# Drop rows where datetime is missing
df.dropna(subset=['Datetime'], inplace=True)

# Fill remaining numeric missing values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Final check
df.isnull().sum()
```

```
Date             0
Time             0
CO(GT)           0
PT08.S1(CO)      0
NMHC(GT)         0
C6H6(GT)         0
PT08.S2(NMHC)    0
NOx(GT)          0
PT08.S3(NOx)     0
NO2(GT)          0
PT08.S4(NO2)     0
PT08.S5(O3)      0
T                0
RH               0
AH               0
Datetime         0
dtype: int64
```

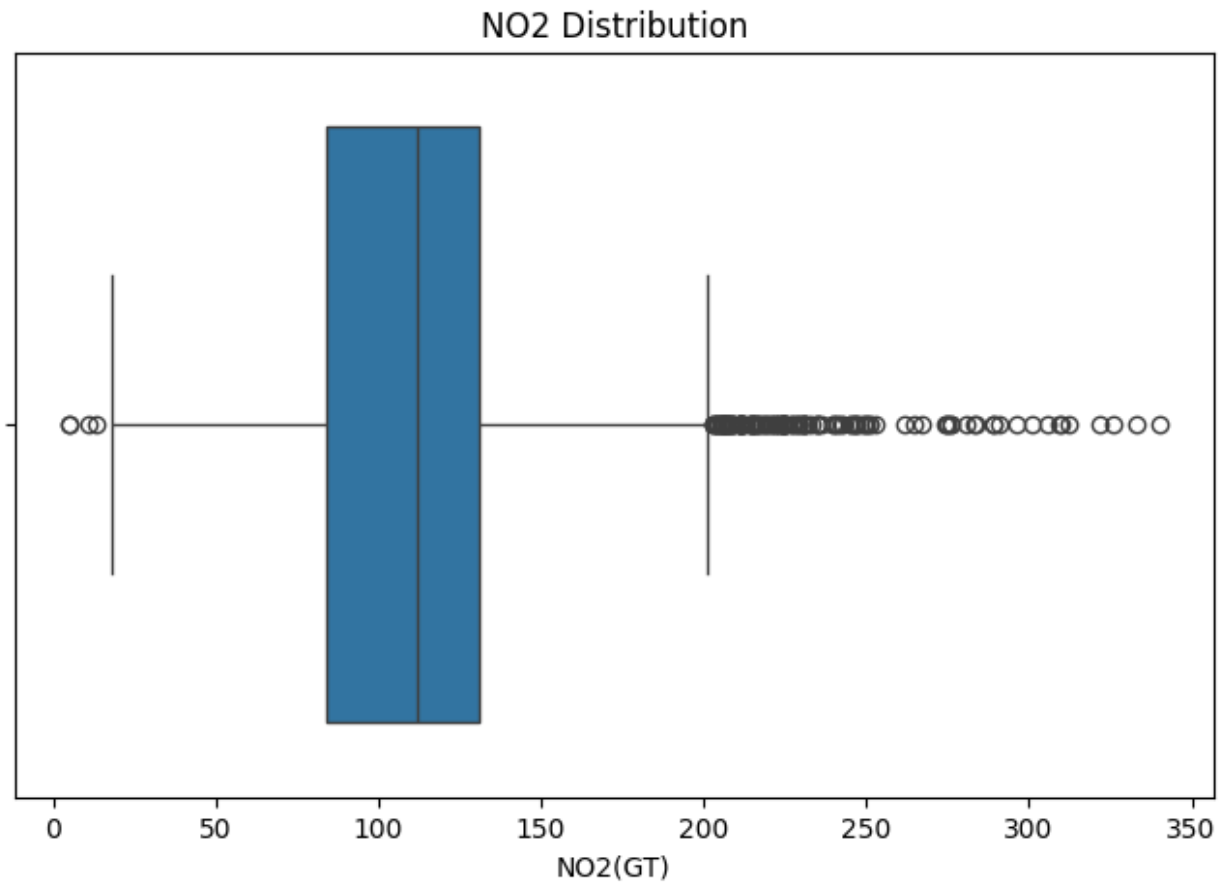
Histogram

```
plt.figure(figsize=(8,5))
plt.hist(df['CO(GT)'], bins=30)
plt.title("CO Distribution")
plt.xlabel("CO(GT)")
plt.ylabel("Frequency")
plt.show()
```



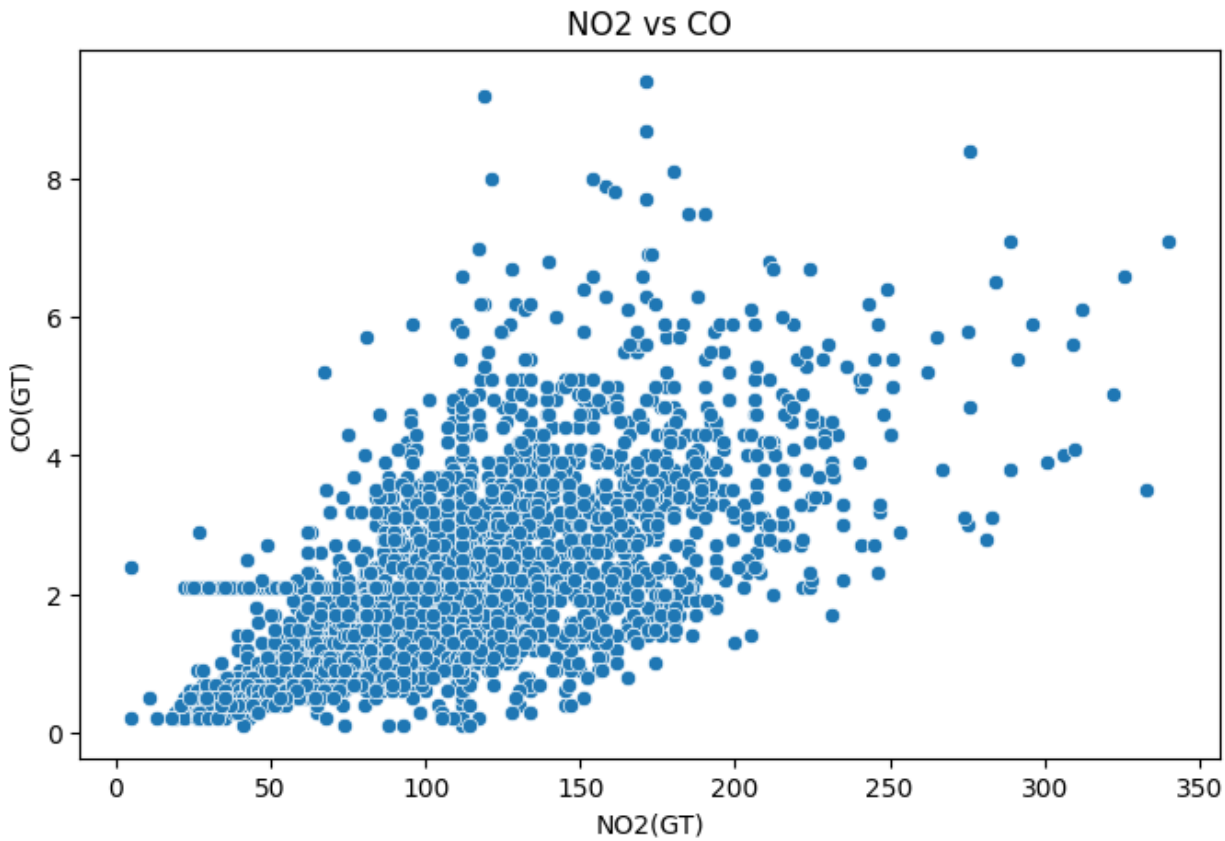
Box Plot

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['NO2(GT)'])
plt.title("NO2 Distribution")
plt.show()
```



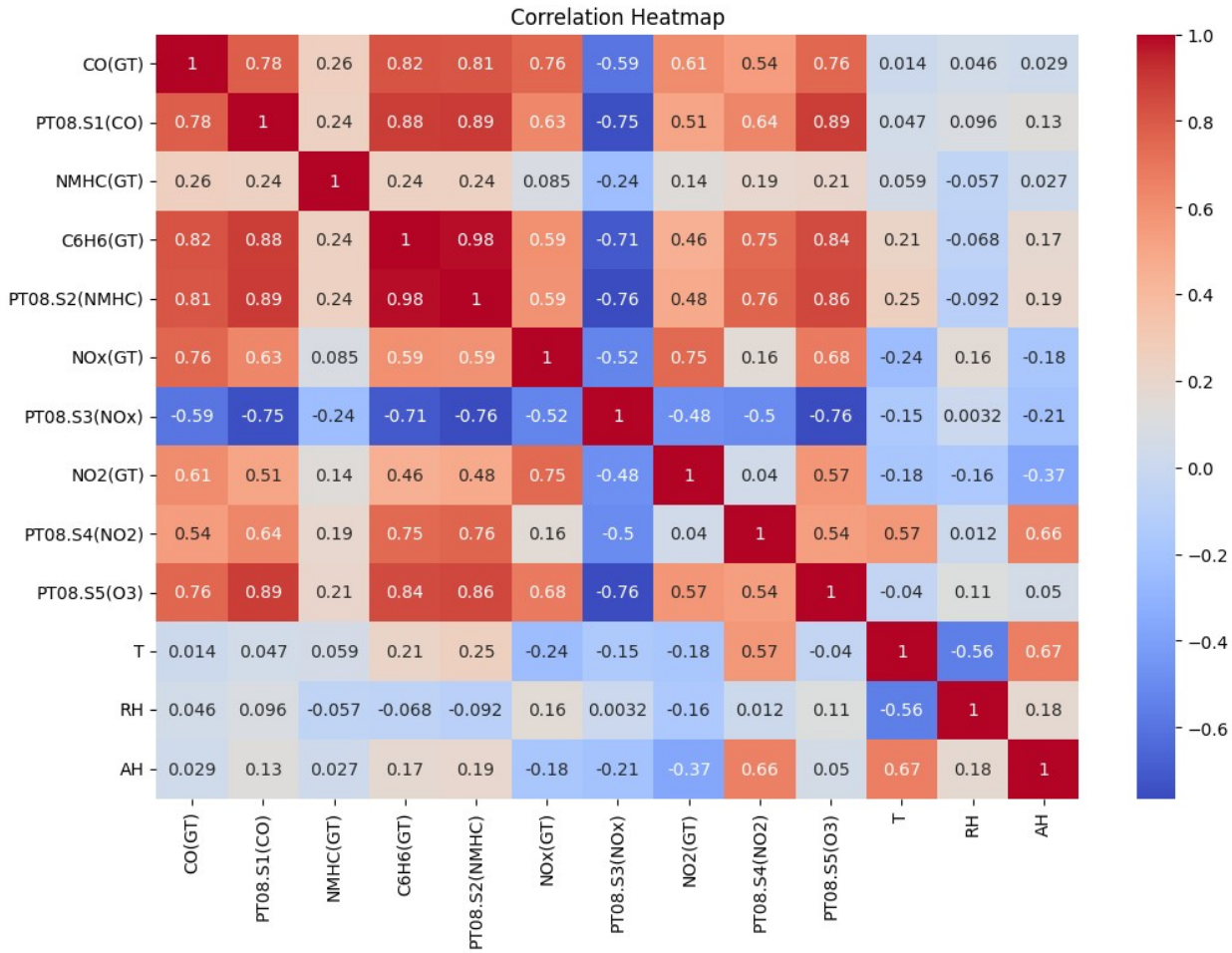
Scatter Plot

```
plt.figure(figsize=(8,5))
sns.scatterplot(x='NO2(GT)', y='CO(GT)', data=df)
plt.title("NO2 vs CO")
plt.show()
```



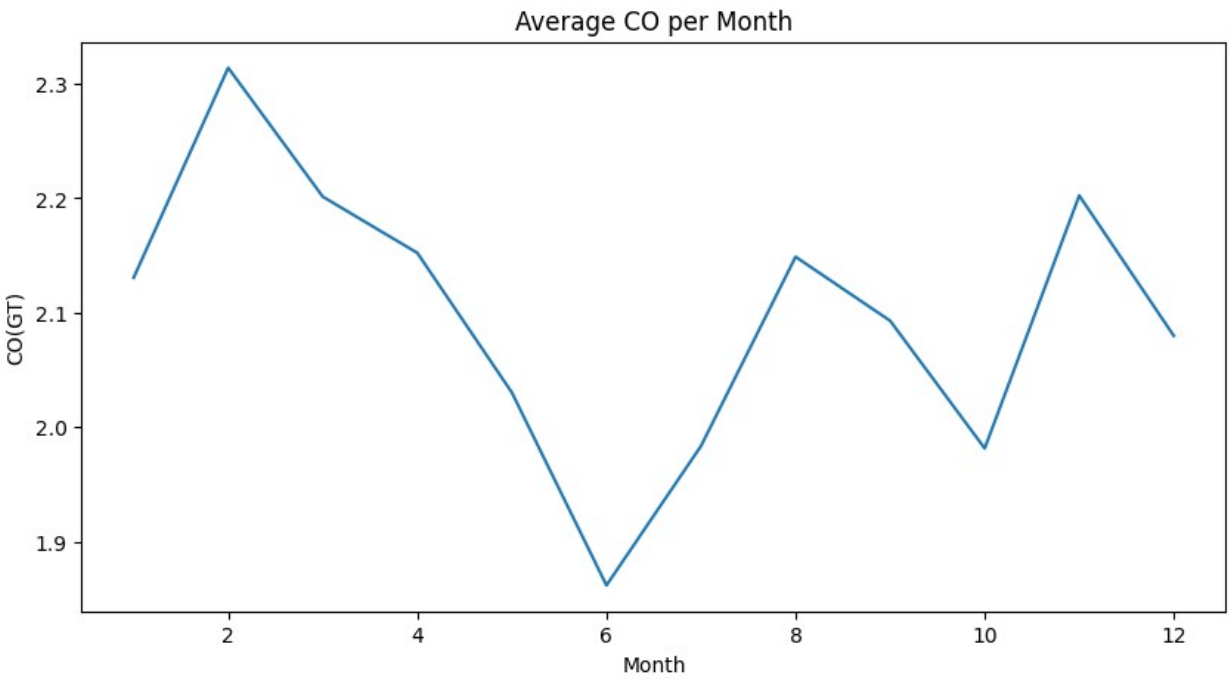
Heatmap

```
plt.figure(figsize=(12,8))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



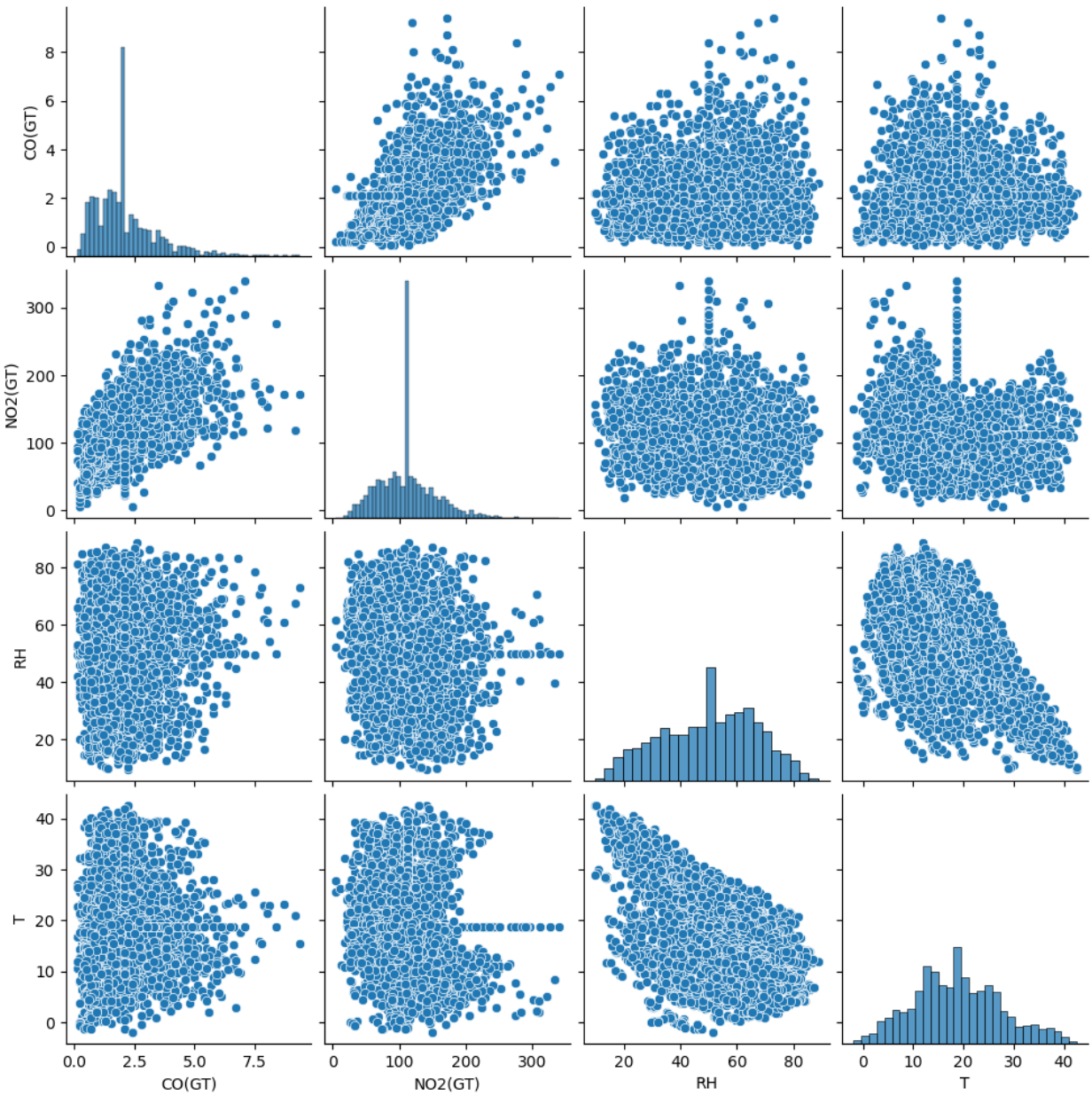
Line Plot

```
df['Month'] = df['Datetime'].dt.month

plt.figure(figsize=(10,5))
df.groupby('Month')['CO(GT)'].mean().plot()
plt.title("Average CO per Month")
plt.xlabel("Month")
plt.ylabel("CO(GT)")
plt.show()
```



Pair Plot

```
sns.pairplot(df[['CO(GT)','NO2(GT)','RH','T']])
plt.show()
```


CO(GT)
NO2(GT)
RH
T
0.0
2.5
5.0
7.5
0
100
200
300
20
40
60
80
0
10
20
30
40